2022

Q1. Write short answers to the following Q:

(a) Discuss Box and Jenkins model building Stages:

The box-Jenkins methodology is a systematic approach for developing Autoregressive Integrated moving Average (ARIMA) Models for time series forecasting. This Process involves several stages that help identify, estimate and validate the model, which are as follows,

1- Identification.

it's purpose is to identify and determine a specific and appropriate model for the time series data.

In this step we check stationarity of time series using plots or statistical tests, and if it's not stationary then differencing is used.

For model selection, we identify Potential ARIMA models (AR, MA, ARMA, or ARIMA) by examining the autocorrelation Function (ACF) and Partial autocorrelation Function (PACF) plotes. these plots help determine the order of Autoregressive (AR), differencing (I) and moving average (MA) components.

2- Estimation:

In this step we estimate the parameters of the identified model.

We can use statistical softwares or methods like Maximum likelihood estimation, least square or Yule walker estimations. (for AR).

3. Diagnostic Checking:

it's purpose is to assess the adequacy of the fitted model.

To check the model Analyze the residuals (diff b/w observed and predicted vals) to ensure they behave like white noise (uncorrelated and N. Distributed).

We perform different statistical tests (such as Ljung-Box Test) to check for the presence of autocorrelation in the residuals.

If the residual exhibits significant autocorrelation then go to the identification stage to refine the model

4. Forecasting.

If the model passes the diagnostic check then the underlying model can be used for Forecasting. we can generate Forecasts based on Final ARIMA Model. it provides confidence intervals for these Forecasts to quantify the uncertainty.

The Box-Jenkins approach is iterative. If the diagnostics indicate that the model is inadequate, the process returns to the identification stage for model refinement. This loop continues until a satisfactory model is found.

b. Explain Portmanteau tests.

The Portmanteau test is a test used to check the overall randomness of residuals in a time series model. portmanteau tests are typically used in the diagnostic checking stage of time series modeling (such as in ARIMA modeling) to validate that the fitted model adequately explains the data

If the test indicates that residuals are not white noise, it suggests the model may require further refinements such as adding more MA or AR terms.

Types

It has two types:

1- Box pierce Test:

It is the original version of portmanteau test, introduced by George Box and David pierce. It examines whether a group of autocorrelations of residuals is significantly diff from zero.

It's test statistic is:

$$Q = n \sum_{k=1}^{m} \hat{\rho}_k^2$$

where;

$n \Rightarrow$ No of obs

$m \Rightarrow$ No of lags considered

$\hat{\rho}_k \Rightarrow$ auto correlation of residuals at lag k.

<u>Null hypothesis</u> $\Rightarrow$ residuals have no significant autocorrelation up to lag m.

<u>Decision Rule</u> $\Rightarrow$

Compare Box pierce test statistic Q to the critical value from chisquare dist with m degrees of freedom, if Q Exceeds the critical value then H_0 is rejected.

2- LJung- Box Test

Purpose ⇒ Tests the Null hypothesis H_0: that the residuals are independently distributed (No autocorrelation) upto a certain lag m.

$$Q = n(n+2) \sum_{k=1}^{m} \frac{\hat{\rho}_k^2}{n-k}$$

Q ⇒ test statistic.

n ⇒ No of obs

m ⇒ Maximum lag being tested.

$\hat{\rho}_k$ ⇒ Sample autocorrelation at lag k.

Decision Rule:

The test statistic Q is compared with or against the critical value from the chi-square distribution with $m-p-q$ degrees of Freedom, where p, q are the orders of the AR and MA terms in the model.

If Q Exceeds the critical value the Null hypothesis is rejected. indicating that the residuals have significat autocorrelation and the models may not be adequate.

C. Define Model selection Criteria AIC and BIC.

AIC stands for Akaike Information Criterion and BIC stands for Byesian Information criterion, which are statistical measures used to compare and select models. they help to balance fit and complexity.

AIC is a criteria used to access the quality of a statistical model by considering both the goodness of fit and complexity of the model.

Formula being used:

$$AIC = -2\ln L + 2k$$

or

$$AIC_{p,q} = \frac{-2\ln(\text{max likelihood})}{n} + 2k$$

$$\approx \ln(\hat{\sigma}_z^2) + k\frac{2}{n} + \text{const}$$

L => Maximum likelihood of the model.

k => No of estimated parameters in the model $(k = p+q+1)$

$\hat{\sigma}_z^2$ => Maximum likelihood estimate of σ_z^2

Interpretations:

A lower AIC value indicates a better model, as it suggests a good fit with fewer parameters.

it penalizes models with more parameters to prevent overfitting.

BIC is another model selection criterion, which like AIC considers model fit and complexity, but with a stronger penality for the No of parameters.

Formula

$$BIC = -2\ln(L) + k\ln(n)$$

or

$$BIC_{p \times q} = \ln(\hat{\sigma}_z^2) + \frac{k\ln(n)}{n}$$

$k = p+q+1$ (No of estimated para)

$n \Rightarrow$ No of obs in data.

$L \Rightarrow$ Maximum likelihood of the model

Interpretation:

A lower BIC value suggest a better model

BIC penalizes more heavily for the No of parameters, making it more comparative / conservative than AIC in selecting complex models.

Comparisons

AIC is more flexible and might favor more complex models, while BIC is strict and tends to select simpler models, especially with large sample sizes.

=> Both criteria are used to select the model that best balances fit, and complexity with the goal of improving out of sample prediction Accuracy.

d. Write a Short Note on Simple Exponential smoothing.

Simpler Exponential smoothing is a forecasting method used for time series data that does not have trends or seasonal patterns. It works by giving more importance to recent data while gradually reducing the influence of older data. The key idea behide SES is to apply exponentially decreasing weights to past observations, with more recent observations recieving higher weights.

Forecasting Formula:

$$\hat{y}_{t+1} = \alpha y_t + (1-\alpha)\hat{y}_{t-1}$$

$\hat{y}_{t+1}$ => Forecast For the next time period.

y_t => Actual value at time t.

$\hat{y}_{t-1}$ => previous Forecast ~~for~~ at time t.

α => Smoothing constant with $0 < \alpha < 1$

selection of α:

The optimal value of α can be chosen by minimizing a criterion such as mean squared error on the forecast errors.

SES calculates the forecast for the next period by combining the most recent actual value and the previous forecast. The balance b/w these two is controlled by a smoothing constant 'α' which ranges from 0 to 1.

A higher α makes the forecast more sensitive to recent changes, while a lower α smooths the data more, making the forecast less reactive to recent changes.

SES is Ideal when you have stable data without trends or seasonal patterns. It's easy to use and needs little data to make accurate forecasts.

It doesn't work well if your data have trends or patterns that repeat over time.

e- Discuss different Forecasting procedure:

Time series analysis and Forecasting are crucial for predicting future trends, behaviors. (based on historical data).
It helps in planning, budgeting, helps businesses to make informed decisions, stock prices, sales fluctuations and strategizing across various domains such as finance, economics, healthcare, etc.

Different Forecasting Procedures involves various methods to predict future values based on past data.
Some commonly used procedures are as follows;

1- Naive forcasting

It is used when data is stable and does not show significant trends or patterns. It is the simplest method where the Forecasts for the next period is Just the value from the current period.

$$\hat{y}_{t+1} = y_t.$$

2- Moving Average:

Averages are fixed No of past observations to smooth out fluctuations, and provide a forecast.

Types:

⇒ simple Moving Average (SMA)

It gives the average of last 'n' observations $\hat{y}_{t+1} = \frac{1}{n}\sum_{i=0}^{n-1} y_{t-i}$

⇒ weighted Moving average (WMA)

Average of the last n obs with diff weights:

$$\hat{y}_{t+1} = \sum_{i=0}^{n-1} w_i y_{t-i}$$

⇒ where w_i are weights that sum up to 1.

Moving Average Forecasts are useful for short term Forecasts when there are no trends or seasonality.

3. Exponential smoothing:

Similar to Moving Average but gives more weight to recent observations through a smoothing constant "α".

It is suitable for various data types depending on the presence of trends and seasonal patterns.

Types:

⇒ Simple exponential smoothing:

weighted average of past obs with exponentially decreasing weights.

$$\hat{y}_{t+1} = \alpha y_t + (1-\alpha)\hat{y}_{t-1}$$

where α is the smoothing constant.

⇒ Holt's linear trend model:

Extends simple Exponential smoothing to account for trends. this method is very useful in agricultural modeling and Forecasting (esp with trend pattern data).

Holt's linear trend method comprises two smoothing constants, two smoothing equations and one Forecast equation:

$\hat{Y}_{t+k} = \alpha_t + k C_t$ Forecast eq.

$\alpha_t = \gamma Y_t + (1-\gamma)(\alpha_{t-1} + C_{t-1})$ Level equation

$C_t = \delta(\alpha_t - \alpha_{t-1}) + (1-\delta) C_{t-1}$ trend eq.

γ, δ ⇒ constants of level and trend of T.S respect. $0 < \gamma, \delta < 1$

Y_t ⇒ denotes observation at time t.

α_t, C_t ⇒ estimates of level and trend of T.S respect.

k ⇒ lags.

⇒ Holt-winters Seasonal Model:

This method has 3 major aspects for performing the Predictions. as it has an average value with the trend and seasonality. it is also known as triple exponential smoothing. It is used for time series data that exhibits both trends and seasonality.

Level

$$\hat{L}_t = \alpha \left(\frac{y_t}{S_{t-k}} \right) + (1-\alpha)(\hat{L}_{t-1} + \hat{T}_{t-1})$$

$\alpha \Rightarrow$ smoothing constant for level

$y_t \Rightarrow$ actual value at time t.

$S_{t-k} \Rightarrow$ is the seasonal component from the same season in the previous cycle.

$\hat{L}_{t-1}$ & $\hat{T}_{t-1} \Rightarrow$ level and trend from prev period.

Trend ($\hat{T}_t$)

$$\hat{T}_t = \beta(\hat{L}_t + \hat{L}_{t-1}) + (1-\beta)\hat{T}_{t-1}$$

$\beta \Rightarrow$ smoothing constant for the trend.

Seasonal Component ($\hat{S}_t$):

$$\hat{S}_t = \gamma \frac{y_t}{\hat{L}_t} + (1-\gamma) \hat{S}_{t-k}$$

$\gamma =$ smoothing constant for seasonal comp

$\hat{S}_{t-m} \Rightarrow$ seasonal component from the same season in prev cycle

$k \Rightarrow$ No of periods in a cycle

Forcasting:

$$\hat{y}_{t+h} = (\hat{L}_t + h.\hat{T}_t) . \hat{S}_{t+h-k}$$

h is the forecast horizon (how far into the future you want to forecast.

4. Autoregressive Integrated Moving Average (ARIMA)

it combines autoregression (AR) differencing (I) and Moving Average (MA) components to model the time series. It is suitable for data with trends and seasonality after appropriate differencing. It's Formula is given by:

$$\phi(B)(1-B)^d y_t = \theta(B) \epsilon_t$$

"$\phi(B)$" & "$\theta(B)$" are polynomials in the back shift operator B.

'd' is the differencing order

ϵ_t is white noise Error.

5. Linear Regression:

it Models the relationship b/w the time series and one or more independent variables. It is mostly used when forecasting is influenced by one or more predictor variables.

It's Formula being used is,

$$y_t = \beta_0 + \beta_1 t + \epsilon_t$$

Q# 2

Find the maximum likelihood estimates of ϕ_1 and ϕ_2 for the given AR(2) Model.

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + Z_t$$

where $Z_t \sim N(0, \sigma_z^2)$

Solution

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + Z_t$$

let $x_1, x_2 \ldots x_n$ be a sample of 'n' observations then.

$$L = f(x_1, x_2, \ldots, x_n)$$

where $Z_t \sim N(0, \sigma_z^2)$

$X_t \sim N(0, \sigma_x^2)$

we have

$$L = f(x_1, x_2) \cdot f(x_3 / x_1, x_2) \cdots f(x_n / x_{n-2} x_{n-1})$$

$$L = f(x_1, x_2) \prod_{t=1}^{n} f(X_t / X_{t-1} X_{t-2})$$

taking ln on B.S:-

$$\ln L = \ln f(x_1, x_2) + \sum_{t=3}^{n} \ln f(X_t / X_{t-1} \cdot X_{t-2}) \quad \text{—— (A)}$$

As AR(2) process is:

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + Z_t$$

$$E(X_t / X_{t-1} \cdot X_{t-2}) = \phi_1 X_{t-1} + \phi_2 X_{t-2}$$

$$V(X_t / X_{t-1} \cdot X_{t-2}) = \sigma_z^2$$

So:

$$(X_t / X_{t-1} \cdot X_{t-2}) \sim N(\phi_1 X_{t-1} + \phi_2 X_{t-2}, \sigma_z^2)$$

$$f(X_t / X_{t-1} \cdot X_{t-2}) = \frac{1}{\sqrt{2\pi\sigma_z^2}} \exp\left[-\frac{1}{2\sigma_z^2} (X_t - \phi_1 X_{t-1} - \phi_2 X_{t-2})\right]^2$$

Taking 'ln' on B.S:

$$\ln f(X_t/X_{t-1} \cdot X_{t-2}) = -\frac{1}{2}\ln(2\pi\sigma_z^2) - \frac{1}{2\sigma_z^2}(X_t - \phi_1 X_{t-1} - \phi_2 X_{t-2})^2$$

$$\ln f(X_t/X_{t-1} \cdot X_{t-2}) = \text{cons} - \frac{1}{2}\ln(\sigma_z^2) - \frac{1}{2\sigma_z^2}(X_t - \phi_1 X_{t-1} - \phi_2 X_{t-2})^2$$

Applying summation:

$$\sum_{t=3}^{n} \ln f(X_t/X_{t-1} \cdot X_{t-2}) = \text{const} - \frac{n-2}{2}\ln\sigma_z^2 - \frac{1}{2\sigma_z^2}\sum_{t=3}^{n}(X_t - \phi_1 X_{t-1} - \phi_2 X_{t-2})^2$$

(B)

Now Joint dist of X_1, X_2

$$(X_1 \cdot X_2) \sim N(0, \Sigma_2)$$

$$f(X_1, X_2) = \frac{1}{(2\pi)^{2/2}|\Sigma_2|^{1/2}} \exp\left[-\frac{1}{2}(X-0)' \Sigma_2^{-1} (X-0)\right].$$

$$f(X_1, X_2) = \frac{1}{(2\pi)\,|\Sigma_2|^{1/2}} \exp\left[-\frac{1}{2}X' \Sigma_2^{-1} X\right]$$

taking ln on B.S:

$$\ln f(X_1, X_2) = \text{const} - \frac{1}{2}\ln|\Sigma_2| - \frac{1}{2}(X' \Sigma_2^{-1} X)$$

as we know:

$$\Sigma_2 = \sigma_z^2 M_2^{-1}$$

$$|\Sigma_2| = |\sigma_z^2 M_2^{-1}|$$

$$= (\sigma_z^2)^2 |M_2|^{-1}$$

$$\ln f(x_1, x_2) = \text{const} - \frac{1}{2}\ln\left((\sigma_z^2)^2 |\underset{\sim}{M_2}|^{-1}\right) - \frac{1}{2}\left(X' \underset{\sim}{\Sigma_2^{-1}} X\right)$$

$$= \text{Const} - \ln \sigma_z^2 + \frac{1}{2}\ln|M_2| - \frac{1}{2}\left(X'(\sigma_z^2 M_2^{-1})^{-1} X\right)$$

$$= \text{Const} - \ln \sigma_z^2 + \frac{1}{2}\ln|M_2| - \frac{1}{2\sigma_z^2}(X' M_2 X) \quad (C)$$

Putting eq B & C in (A).

$$\ln L = \ln f(x_1 x_2) + \sum_{t=3}^{n} \ln f(x_t / x_{t-1} x_{t-2})$$

$$= \text{Const} - \frac{n}{2}\ln\sigma_z^2 + \ln\sigma_z^2 - \frac{1}{2\sigma_z^2}\sum_{t=3}^{n}(X_t - \phi_1 X_{t-1}$$

$$- \phi_2 X_{t-2})^2 + \text{Cons} - \frac{n}{2}\ln\sigma_z^2 + \frac{1}{2}\ln|M_2| - \frac{1}{2\sigma_z^2}$$

$$\left[\sum_{t=3}^{n}(X_t - \phi_1 X_{t-1} - \phi_2 X_{t-2})^2 + X' \underset{\sim}{M_2} X\right]$$

To find MLE's of ϕ_1, ϕ_2, we partially differentiate eq (D) w.r.t ϕ_1 and ϕ_2. Assuming that 'n' is large and we are ignoring $\frac{1}{2}\ln(M_2)$ and $\underline{X' \underset{\sim}{M_2} X}$

$$\frac{\partial \ln L}{\partial \phi_1} = 0 - 0 + 0 - \frac{1}{2\sigma_z^2}\left[2\sum_{t=3}^{n}(X_t - \phi_1 X_{t-1} - \phi_2 X_{t-2})(-X_{t-1})\right]$$

equating to zero.

$$\frac{1}{\sigma_z^2}\left[\sum_{t=3}^{n}(X_t - \phi_1 X_{t-1} - \phi_2 X_{t-2})(X_{t-1})\right] = 0$$

$$\frac{1}{\sigma_z^2}\left[\sum_{t=3}^{n}(X_t-\phi_1 X_{t-1}-\phi_2 X_{t-2})(X_{t-1})\right]=0$$

$$\sum_{t=3}^{n} X_t X_{t-1}-\hat{\phi}_1\sum_{t=3}^{n} X_{t-1}^2-\hat{\phi}_2\sum_{t=3}^{n} X_{t-2}X_{t-1}=0$$

Dividing by $\sum_{t=3}^{n} X_{t-1}^2$

$$\frac{\sum_{t=3}^{n} X_t X_{t-1}}{\sum_{t=3}^{n} X_{t-1}^2}-\hat{\phi}_1\frac{\cancel{\sum_{t=3}^{n} X_{t-1}^2}}{\cancel{\sum_{t=3}^{n} X_{t-1}^2}}-\frac{\hat{\phi}_2\sum_{t=3}^{n} X_{t-2}X_{t-1}}{\sum_{t=3}^{n} X_{t-1}^2}=0$$

$\div$ by

$$\frac{\sum_{t=3}^{n} X_t X_{t-1}/n-2}{\sum_{t=3}^{n} X_{t-1}^2/n-2}-\hat{\phi}_1-\frac{\hat{\phi}_2\sum_{t=3}^{n} X_{t-2}X_{t-1}/n-2}{\sum_{t=3}^{n} X_{t-1}^2/n-2}=0$$

as n is large;

$$\frac{C_1}{C_0}-\hat{\phi}_1-\hat{\phi}_2\frac{C_1}{C_0}=0$$

$$r_1-\hat{\phi}_1-\hat{\phi}_2 r_1=0$$

$$\hat{\phi}_1=r_1-\hat{\phi}_2 r_1$$

$$\hat{\phi}_1=r_1(1-\hat{\phi}_2) \quad \text{—(i)}$$

Similarly partially diff w.r.t $\hat{\phi}_2$:

$$\frac{\partial \ln L}{\partial \hat{\phi}_2}=\frac{-1}{2\sigma_z^2}\left[2\sum_{t=3}^{n}(X_t-\phi_1 X_{t-1}-\phi_2 X_{t-2})(-X_{t-2})\right]$$

equating to zero;

$$\frac{1}{\sigma_z^2}\left(\sum_{t=3}^{n}(X_t-\phi_1 X_{t-1}-\phi_2 X_{t-2})(X_{t-2})\right]=0$$

$$\sum_{t=3}^{n} X_t X_{t-2} - \phi_1 \sum_{t=3}^{n} X_{t-1} X_{t-2} - \phi_2 \sum_{t=3}^{n} X_{t-2}^2 = 0.$$

Dividing by $\sum_{t=3}^{n} X_{t-2}^2$

$$\frac{\sum_{t=3}^{n} X_t X_{t-2}}{\sum_{t=3}^{n} X_{t-2}^2} - \phi_1 \frac{\sum_{t=3}^{n} X_{t-1} X_{t-2}}{\sum_{t=3}^{n} X_{t-2}^2} - \phi_2 \cancel{\frac{\sum_{t=3}^{n} X_{t-2}^2}{\sum_{t=3} X_{t-2}^2}} = 0$$

$$\frac{\sum_{t=3}^{n} X_t X_{t-2}/n-2}{\sum_{t=3}^{n} X_{t-2}^2/n-2} - \phi_1 \frac{\sum_{t=3}^{n} X_{t-1} X_{t-2}/n-2}{\sum_{t=3}^{n} X_{t-2}^2/n-2} - \phi_2 = 0$$

$$\frac{C_2}{C_0} - \phi_1 \frac{C_1}{C_0} - \phi_2 = 0$$

$$r_2 - \hat{\phi}_1 r_1 - \hat{\phi}_2 = 0$$

using $\hat{\phi}_1 = r_1(1-\hat{\phi}_2)$

$$r_2 - r_1(r_1(1-\hat{\phi}_2)) - \hat{\phi}_2 = 0$$

$$r_2 - r_1^2 + r_1^2\hat{\phi}_2 - \hat{\phi}_2 = 0$$

$$r_2 - r_1^2 = \hat{\phi}_2 - r_1^2\hat{\phi}_2$$

$$r_2 - r_1^2 = \hat{\phi}_2(1-r_1^2)$$

$$\frac{r_2 - r_1^2}{(1-r_1^2)} = \hat{\phi}_2$$ required estim[ate] of ϕ_2

As $\hat{\phi}_1 = r_1(1-\hat{\phi}_2)$

$$= r_1\left(1 - \frac{r_2 - r_1^2}{1-r_1^2}\right)$$

$$= r_1\left[\frac{1-\cancel{r_1^2} - r_2 + \cancel{r_1^2}}{1-r_1^2}\right]$$

$$= r_1\left(\frac{1-r_2}{1-r_1^2}\right)$$

$$\hat{\phi}_1 = \frac{r_1 - r_1 r_2}{1-r_1^2}$$ => required estimate of ϕ_1

Q#3:-

Find the minimum mean square Error Forecast X_{n+1} for AR(1) Process with zero mean using MA Representation. Also Find it's mean and variance.